

XLOOKUP Basic

See details →

[illegible]

XLOOKUP Basic Explanation



What I Did

Looked up Total Sale by Order ID, Customer Name by Cust ID, and tested a custom fallback message for a missing ID.



How It Works

XLOOKUP scans the lookup_array for the value, then returns the matching cell from return_array. The 4th argument defines what to show when no match is found.



Key Insight

Setting the 4th arg to "Not Found" replaces the default #N/A error with a readable message.

Formula Used

```
=XLOOKUP(J4, B3:B12,  
H3:H12, "", 0)
```

```
=XLOOKUP(J8, D3:D12,  
C3:C12, "", 0)
```

```
=XLOOKUP(J12, D3:D12,  
E3:E12, "Not Found", 0)
```


XLOOKUP - All Data Explanation



What I Did

Retrieved every field of an order record (all 6 columns) using a single Order ID as the lookup key.



How It Works

By passing a multi-column range (C3:H12) as the return_array, XLOOKUP spills the entire matched row automatically.



Key Insight

No need to write separate formulas per column, XLOOKUP spills all fields at once.

Formula Used

```
=XLOOKUP (B15, B3:B12,  
          C3:H12, "", 0)
```

XLOOKUP - Multiple Records Match

See details →

[illegible]

XLOOKUP - Multiple Records Explanation



What I Did

Matched rows using two criteria simultaneously (e.g. Order ID + Customer Name, or Cust ID + Product Name).



How It Works

Concatenating the lookup value and array columns creates a single virtual key column, enabling multi-condition lookup without helper columns.



Key Insight

The & operator joins two arrays into one, a clean multi-criteria trick with no extra columns needed.

Formula Used

```
=XLOOKUP (J4&K4,  
  $B$3:$B$11&$C$3:$C$11,  
  $D$3:$H$11)
```

```
=XLOOKUP (J9&K9,  
  $D$3:$D$11&$E$3:$E$11,  
  F3:H11)
```

XLOOKUP + TRANSPOSE

[See details →](#)

	A	B	C	D	E	F	G	H	I	J	K	
1	XLOOKUP + Transpose											
2												
3		Order ID	Customer Name	Cust ID	Product Name	Qty	Unit Price	Total Sale		Order ID	ORD011	
4		ORD001	Rahim Uddin	CU01	Headphones	2	1380	2760		Customer Name	Anwar Hossain	
5		ORD002	Karim Hossain	CU02	Chair	2	5848	11697		Cust ID	CU11	
6		ORD003	Salma Akter	CU03	Table	1	15208	15208		Product Name	Chair	
7		ORD004	Anika Begum	CU04	USB Cable	2	156	313		Qty	2	
8		ORD005	Tariq Khan	CU05	Charger	3	280	840		Unit Price	156	
9		ORD006	Farhan Ahmed	CU06	Headphones	4	1380	5520		Total Sale	313	
10		ORD007	Nabila Rahman	CU07	Tablet	4	27025	108100				
11		ORD008	Rafiq Islam	CU08	Power Bank	1	1400	1400				
12		ORD009	Sabina Yasmin	CU09	Charger	1	280	280				
13		ORD010	Jamal Uddin	CU10	Headphones	1	15208	15208				
14		ORD011	Anwar Hossain	CU11	Chair	2	156	313				
15		ORD012	Nusrat Jahan	CU12	Table	3	280	840				
16		ORD013	Tanvir Ahmed	CU13	USB Cable	4	1380	5520				
17		ORD014	Shabnam Akter	CU14	Charger	4	27025	108100				

XLOOKUP + TRANSPOSE Explanation



What I Did

Fetches a full order row and rotated it vertically, turning the horizontal result into a neat record card layout.



How It Works

XLOOKUP returns a horizontal array. Wrapping it in TRANSPOSE flips rows and columns, displaying each field on its own row.



Key Insight

TRANSPOSE works on spilled arrays, combine it with XLOOKUP to build vertical profile cards instantly.

Formula Used

```
=TRANSPOSE (  
  XLOOKUP (K3,  
    $B$3:$B$19,  
    $C$3:$H$19)  
)
```


Multiple Columns - CHOOSE Explanation



What I Did

Retrieved 3 non-adjacent columns (Customer Name, Product Name, Total Sale) for a given Order ID in one formula.



How It Works

CHOOSE({1,2,3}, col1, col2, col3) builds a virtual 3-column array from any columns you pick, which XLOOKUP then searches and returns.



Key Insight

CHOOSE with an array constant {1,2,3} lets you hand-pick any non-adjacent columns as the return range.

Formula Used

```
=XLOOKUP (B16, B3:B12,  
  CHOOSE ({1, 2, 3},  
    C3:C12,  
    E3:E12,  
    H3:H12) )
```

Multiple Columns - CHOOSECOLS

[See details →](#)

	A	B	C	D	E	F	G	H
1	XLOOKUP Multiple Columns (with choosecols)							
2								
3		Order ID	Customer Name	Cust ID	Product Name	Qty	Unit Price	Total Sale
4		ORD001	Rahim Uddin	CU01	Headphones	2	2500	5000
5		ORD002	Karim Hossain	CU02	Chair	2	6000	12000
6		ORD003	Salma Akter	CU03	Table	1	15000	15000
7		ORD003	Anika Begum	CU04	USB Cable	2	150	300
8		ORD005	Tariq Khan	CU05	Charger	3	500	1500
9		ORD006	Farhan Ahmed	CU06	Headphones	4	2500	10000
10		ORD007	Nabila Rahman	CU07	Tablet	4	20000	80000
11		ORD008	Rafiq Islam	CU08	Power Bank	1	1500	1500
12		ORD009	Sabina Yasmin	CU09	Charger	1	500	500
13								
14								
15		Order ID	Customer Name	Product Name	Total Sale			
16		ORD005	Tariq Khan	Charger	1500			
17								
18		Cust ID	Customer Name	Product Name	Total Sale			
19		CU03	Salma Akter	Table	15000			
20								

Multiple Columns - CHOOSECOLS Explanation



What I Did

Extracted columns 1, 3, and 6 from a full-row XLOOKUP result - once by Order ID and once by Cust ID.



How It Works

CHOOSECOLS selects specific column positions from an array. It wraps the XLOOKUP result to keep only the columns you need.



Key Insight

CHOOSECOLS is cleaner than CHOOSE when your return range is contiguous, just pass column position numbers.

Formula Used

```
=CHOOSECOLS (  
  XLOOKUP (B16, B3:B12,  
    C3:H12), 1, 3, 6)
```

```
=CHOOSECOLS (  
  XLOOKUP (B19, D3:D12,  
    C3:H12), 1, 3, 6)
```


Two-Way XLOOKUP Explanation



What I Did

Found the value at the intersection of a specific row (Order ID) and a specific column (e.g. Total Sale) - true 2D lookup.



How It Works

The inner XLOOKUP finds the matching column range by header name. The outer XLOOKUP uses that result to find the right row.



Key Insight

Nesting XLOOKUP in the return_array argument turns it into a dynamic 2D intersection formula.

Formula Used

```
=XLOOKUP (B15,  
  B3:B12,  
  XLOOKUP (C14,  
    B3:H3,  
    B3:H12) )
```


Nested XLOOKUP - Multiple Tables Explanation



What I Did

Searched for an Order ID across 3 separate data tables, if not found in table 1, it automatically tried table 2, then table 3.



How It Works

Each XLOOKUP is placed in the 4th (if_not_found) argument of the previous one, chaining the searches into a waterfall logic.



Key Insight

The if_not_found argument is itself an XLOOKUP, this chains table searches without IF or IFERROR.

Formula Used

```
=XLOOKUP (B22,  
  B3:B7, C3:H7,  
  XLOOKUP (B22,  
    B9:B13, C9:H13,  
    XLOOKUP (B22,  
      B15:B18,  
      C15:H18) ) )
```


	A	B	C	D	E	F	G	
1	XLOOKUP + TRIM							
2								
3		Order ID	Customer Name	Cust ID	Product Name	Qty	Unit Price	
4		ORD001	Manish Jain	CU01	Headphones	2	1380	
5		ORD002	Priya Singh	CU02	Chair	2	5848	
6		ORD003	Rahul Mehta	CU03	Table	1	15208	
7		ORD003	Anjali Gupta	CU04	USB Cable	2	156	
8		ORD005	Pooja Rao	CU05	Charger	3	280	
9		ORD006	Kiran Joshi	CU06	Headphones	4	1380	
10		ORD007	Sneha Verma	CU07	Tablet	4	27025	
11		ORD008	Priya Singh	CU08	Power Bank	1	1400	
12		ORD009	Manish Jain	CU09	Charger	1	280	
13								
14		Order ID	Customer Name					
15		ORD007	Sneha Verma					
16								

XLOOKUP + TRIM Explanation



What I Did

Handled dirty source data where Order IDs had invisible leading or trailing spaces that caused lookup failures.



How It Works

Wrapping the lookup_array inside TRIM() cleans all whitespace on the fly during the lookup — no need to fix the source data.



Key Insight

TRIM inside XLOOKUP is a real-world data cleaning technique — the source data stays untouched.

{ } Formula Used

```
=XLOOKUP (B15,  
  TRIM (B3:B12) ,  
  C3:C12)
```

XLOOKUP - Phone Number Lookup

See details →

	A	B	C	D	E	F	G	H	
25	Full_Name	Phone_Number_ Case 2	Gender	DOB	Family_Member_Qty	City	Division	Registration_Date	
26	Rahim Uddin	122285197	Male	28667	1	Rangpur Sadar	Rangpur	43958	
27	Karim Hossain	159679341	Male	26842	3	Subidbazar	Sylhet	44037	
28	Salma Akter	125575449	Female	37427	5	Shah Makhdum	Rajshahi	44130	
29	Anika Begum	114884456	Female	31587	4	Kawran Bazar	Barishal	45259	
30	Tariq Khan	191269394	Male	29762	1	Gollamari	Khulna	44279	
31	Farhan Ahmed	112904003	Male	32317	3	Boalia	Rajshahi	43923	
32	Nabila Rahman	136117555	Female	25382	5	Gollamari	Khulna	45006	
33	Rafiq Islam	134826513	Male	25017	3	Dhanmondi	Dhaka	45129	
34	Sabina Yasmin	126628927	Female	34872	6	Kawran Bazar	Barishal	44925	
35	Jamal Uddin	0174321652	Male	26842	1	Muktagachha	Mymensingh	45064	
36	Anwar Hossain	0138263100	Male	28667	5	Panchlaish	Chattogram	45224	
37	Nusrat Jahan	0118198675	Female	29397	3	Panchlaish	Chattogram	43982	
38	Tanvir Ahmed	0113057825	Male	27572	5	Sonadanga	Khulna	43885	
39	Shabnam Akter	0183046040	Female	33777	2	Gollamari	Khulna	44085	
40	Imran Karim	0167125592	Male	38522	5	Dhanmondi	Dhaka	45131	
41	Ayesha Siddique	0193295518	Female	26842	5	Panchlaish	Chattogram	44217	
42	Hasan Mahmud	0123623128	Male	39617	5	Kaunia	Rangpur	45072	
43	Lutfun Nahar	0144885857	Male	37792	3	Bakerganj	Barishal	45010	
44	Rashed Chowdhury	0119395818	Male	22827	4	Godagari	Rajshahi	44076	
45	Farida Parveen	0139639203	Female	30492	1	Shah Makhdum	Rajshahi	44632	
46									

XLOOKUP - Phone Number Lookup Explanation



What I Did

Looked up full customer details using only the last 9 digits of a phone number, bypassing country code format differences.



How It Works

RIGHT() extracts the last 9 characters from both the search value and the phone column, so +880 prefixes are ignored during matching.



Key Insight

Applying RIGHT() inside both lookup arguments makes the formula resilient to inconsistent phone number formats.

Formula Used

```
=XLOOKUP(  
  RIGHT(B26, 9),  
  RIGHT($I$1:$I$21, 9),  
  $C$1:$H$21)
```